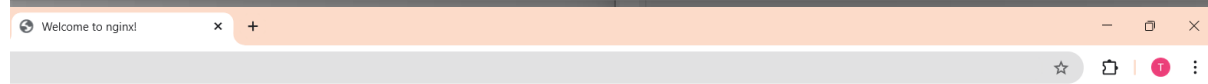


Bài 1: Chạy một container đơn giản với Docker Compose

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 1>docker compose up -d
[+] up 1/1
✔Container nginx-bail Started

T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 1>docker compose ps
NAME          IMAGE      COMMAND                  SERVICE    CREATED      STATUS      PORTS
nginx-bail    nginx     "/docker-entrypoint...." web        30 seconds ago Up 7 seconds 80/tcp

T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 1>
```



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to fs.com/nginx.

Thank you for using nginx.

Bài 2: Chạy MySQL với Docker Compose

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 2>docker compose up -d
[+] up 16/16
✔Image mysql:8.0 Pulled
✔Network bi2_default Created
✔Container mysql-bai2 Started
```

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 2>docker compose exec db mysql -u user -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.45 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |
```

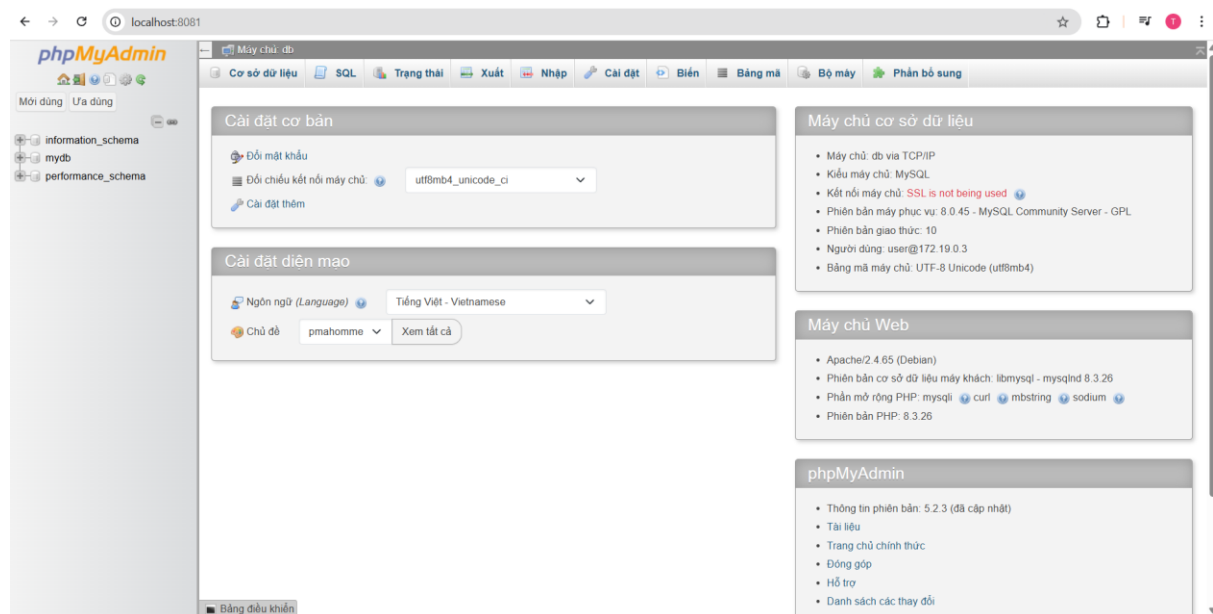
```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mydb      |
| performance_schema |
+-----+
3 rows in set (0.02 sec)

mysql> |
```

Bài 3: Kết nối MySQL với PHPMYAdmin

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 3>docker compose up -d
[+] up 26/26
✓Image phpmyadmin/phpmyadmin Pulled 656.3s
✓Network bi3_default Created 0.2s
✓Volume bi3_mysql_data Created 0.0s
✓Container mysql-bai3 Started 2.4s
✓Container phpmyadmin-bai3 Started 1.3s

T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 3>docker compose logs -f
mysql-bai3 | 2026-03-18 13:22:54+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.45-1.el9 starte
d.
```



Bài 4: Chạy ứng dụng Node.js với Docker Compose

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 4>docker compose up --build
#1 [internal] load local bake definitions
#1 reading from stdin 599B done
#1 DONE 0.0s
```

```
#12 resolving provenance for metadata file
#12 DONE 0.0s
[+] up 3/3
✓Image bi4-app Built
✓Network bi4_default Created
✓Container node-bai4 Created
Attaching to node-bai4
node-bai4 | Server running on port 3000
|
V View in Docker Desktop O View Config W Enable Watch d Detach
```



Bài 5: Chạy Redis với Docker Compose

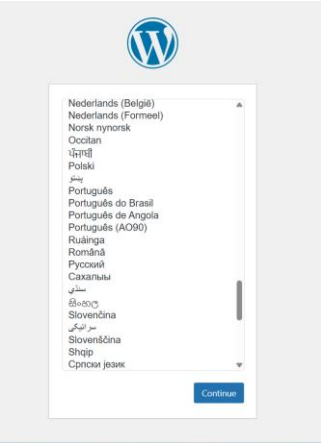
```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 5>docker-compose up -d
time="2026-03-18T20:33:02+07:00" level=warning msg="T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\B
ài 5\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential c
onfusion"
[+] up 13/13
  ✔Image redis:7 Pulled 91.6s
  ✔Network bi5_default Created 0.1s
  ✔Container my-redis Started 0.7s
```

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 5>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
4435c786adb   redis:7   "docker-entrypoint.s..." 7 seconds ago  Up 7 seconds  0.0.0.0:6379->6379/tcp, [::]:6379->6379
/tcp my-redis
```

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 5>docker exec -it my-redis redis-cli
127.0.0.1:6379> ping
PONG
127.0.0.1:6379> |
```

Bài 6: Chạy WordPress với MySQL

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 6>docker-compose up -d
time="2026-03-18T20:49:59+07:00" level=warning msg="T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\B
ài 6\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential
confusion"
[+] up 2/2
  ✔Container wp-mysql Started 0.3s
  ✔Container wp-app Started 0.3s
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 6>
```



Bài 7: Chạy MongoDB với Docker Compose

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 7>docker-compose up -d
time="2026-03-18T21:02:27+07:00" level=warning msg="T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\B
ài 7\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential
confusion"
[+] up 5/7
  ✔Image mongo:6 Pulled 105.4s
  ✔Network bi7_default Created 0.3s
  ✔Volume bi7_mongo_data Created 0.0s
  ✔Container my-mongo Started 2.7s
  ✔Container mongo-express Started 2.2s
```

```
T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 7>docker exec -it my-mongo mongosh
Current Mongosh Log ID: 69bab24a1c215843a88de665
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.10
Using MongoDB:      6.0.27
Using Mongosh:      2.5.10

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

test> use admin
switched to db admin
admin> db.auth("admin", "123456")
{ ok: 1 }
admin> |
```

Bài 8: Kết nối nhiều dịch vụ với Docker Compose

```
mysql_db | 2026-03-22T07:16:22.471532Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.45' socket:
'/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
Container mysql_db Healthy
node_app | 🚀 Server chạy tại port 3000
node_app | ✅ Đã kết nối MySQL
[]

View in Docker Desktop View Config Enable Watch Detach
Ln 41, Col 4 Spaces: 2 UTF-8 CRLF {} JavaScript Signed out Go Live
```

localhost:3000

Pretty print ☐

```
[{"time":"2026-03-22T07:16:26.000Z"}]
```

Bài 9: Chạy ứng dụng Python Flask với Docker Compose

```
#13 resolving provenance for metadata file
#13 DONE 0.1s
[+] up 3/3
✓ Image bi9-flask Built 125.3s
✓ Network bi9_default Created 0.1s
✓ Container flask_app Created 0.2s
Attaching to flask_app
flask_app | * Serving Flask app 'app'
flask_app | * Debug mode: off
flask_app | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI se
rver instead.
flask_app | * Running on all addresses (0.0.0.0)
flask_app | * Running on http://127.0.0.1:5000
flask_app | * Running on http://172.19.0.2:5000
flask_app | Press CTRL+C to quit
flask_app | 172.19.0.1 - - [22/Mar/2026 07:24:47] "GET /hello/Truc HTTP/1.1" 200 -
flask_app | 172.19.0.1 - - [22/Mar/2026 07:24:47] "GET /favicon.ico HTTP/1.1" 404 -
[]

View in Docker Desktop View Config Enable Watch Detach
```

localhost:5000/hello/Truc

Pretty print ☐

```
{"message":"Hello Truc"}
```

Bài 10: Lưu trữ dữ liệu với Docker Volumes

```

PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 10> docker-compose up -d
[+] up 3/3
✓ Network bi10_default      Created
✓ Volume bi10_mysql_data    Created
✓ Container mysql_volume_db Started
PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 10> docker exec -it mysql_volume_db mysql -u root
-p
Enter password:
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)
PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 10> docker exec -it mysql_volume_db mysql -u root
-p
Enter password:
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)
PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 10> docker exec -it mysql_volume_db mysql -u root
-p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.45 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show db
-> ^C

```

Bài 11: Chạy dịch vụ Postgres với Adminer

```

PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 11> docker-compose up -d
[+] up 39/39
✓ Image adminer      Pulled
✓ Image postgres:15  Pulled
✓ Network bi11_default Created
✓ Volume bi11_pg_data Created
✓ Container postgres_db Started
✓ Container adminer_ui Started
PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 11> 

```

localhost:8083

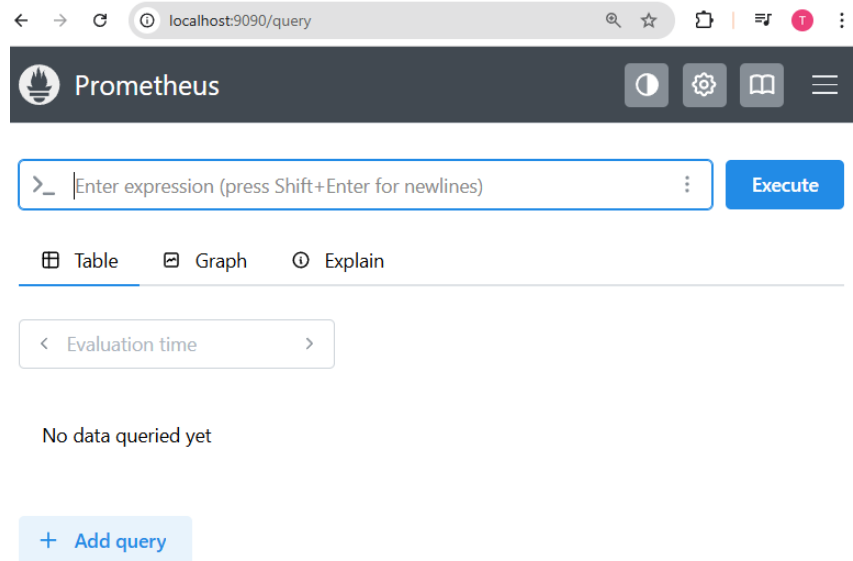
Đăng nhập

Hệ thống	MySQL / MariaDB ▼
Máy chủ	db
Tên người dùng	
Mật khẩu	
Cơ sở dữ liệu	

☐ Giữ đăng nhập một thời gian

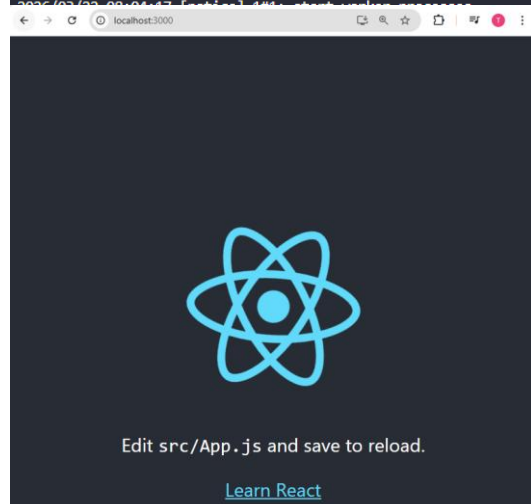
Bài 12: Giám sát container với Prometheus và Grafana

```
PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 12> docker-compose up -d
[+] up 32/32
✓ Image prom/node-exporter Pulled
✓ Image prom/prometheus Pulled
✓ Image grafana/grafana Pulled
✓ Network bi12_default Created
✓ Volume bi12_grafana_data Created
✓ Container node_exporter Started
✓ Container grafana Started
✓ Container prometheus Started
PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 12>
```



Bài 13: Chạy ứng dụng React với Nginx

```
PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 13\react-nginx> docker run -p 3000:80 react-nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: /etc/nginx/conf.d/default.conf differs from the packaged version
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
2026/03/22 08:04:17 [notice] 1#1: nginx/1.29.6
2026/03/22 08:04:17 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/03/22 08:04:17 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/22 08:04:17 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
```



Bài 14: Cấu hình mạng riêng giữa các container

```
PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 14> docker-compose up -d
[+] up 7/7
✓ Image busybox          Pulled
✓ Image nginx:alpine     Pulled
✓ Network bi14_mynetwork Created
✓ Container app2         Started
✓ Container app1         Started
PS T:\Kiến trúc máy tính\Thực hành\Tuan05\Docker Compose\Phần 2\Bài 14> docker exec -it app2 sh
/ # ping app1
PING app1 (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=2.658 ms
```

Bài 15: Giới hạn tài nguyên cho container

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
21ac2789794d	redis_stack_redis.1.73jlhdc6or4e6edj5oblsgtg	0.17%	6.719MiB / 200MiB	3.36%	914B / 378B	14.8MB / 0B	6